

SKU Architecture Compendium

Nine SKU architecture sheets plus the D100 drone SoC and the DG SDV platform — each sheet carries its block architecture, timing behaviour, characteristic curves, market sizing (flagged as internal estimates) and the FPGA-prototype → 130nm-ASIC boundary. Built to survive a semiconductor-literate diligence read.

BLDC Motor Controller — AEC-Q100 · ISO 26262 ASIL-B/C

SKU - 1

Per the product brief: DGridRiscV 200 MHz · HW PID <1 μ s · FOC datapath · PWM×7 · 5–120 V B/B rail · ReRAM 256KB–1MB.

Replaces

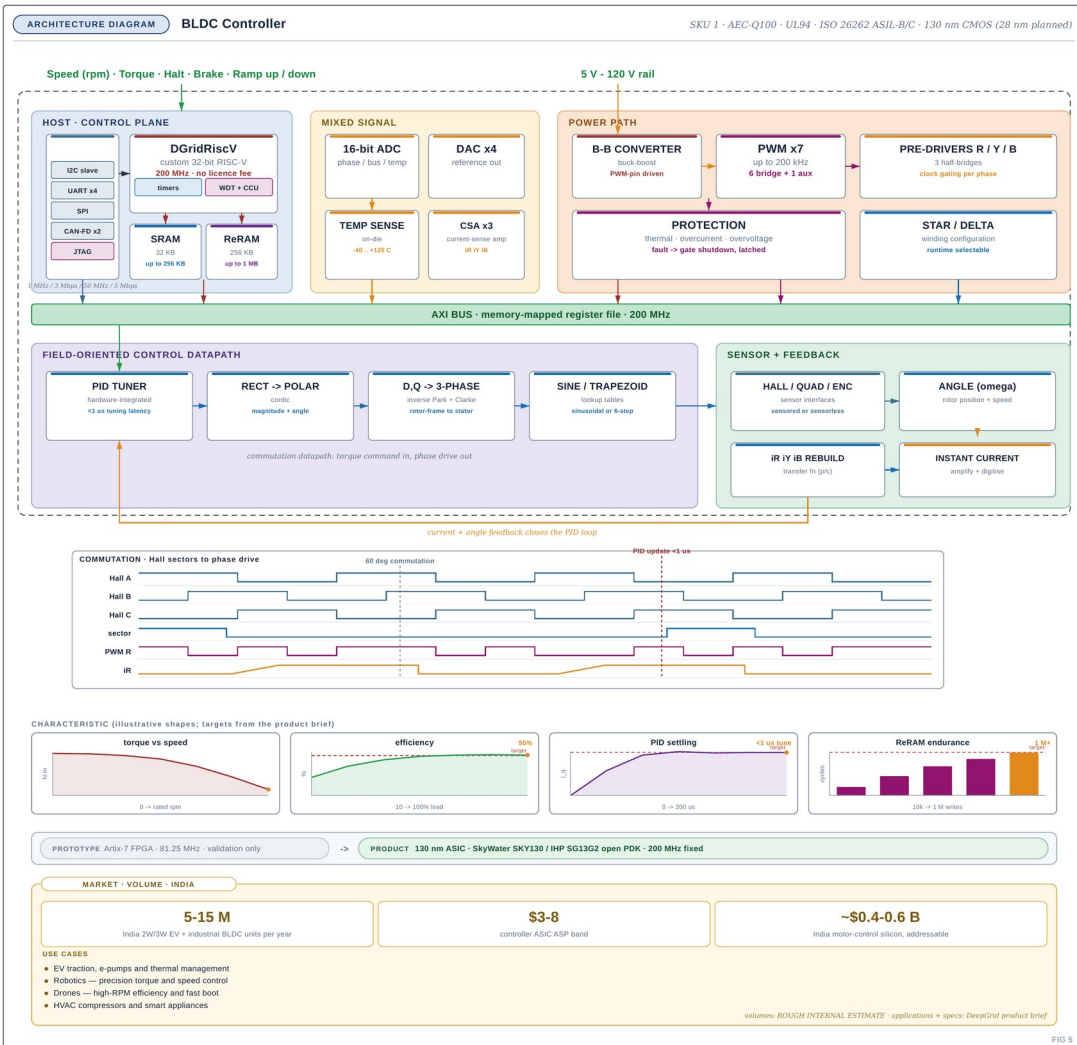
DRV83xx-class driver + external MCU two-chip sets in fans, appliances, EV 2W/3W, robotics and ASWA actuator joints — collapsed to one die with runtime star/delta.

Socket & policy hook

Captive: every ASWA joint and D100 module. Commercial: BEE star-rating push to BLDC · ₹250–550 Cr/yr fans alone. Sheet market panel flagged as internal estimate — kept honest.

Status & node path

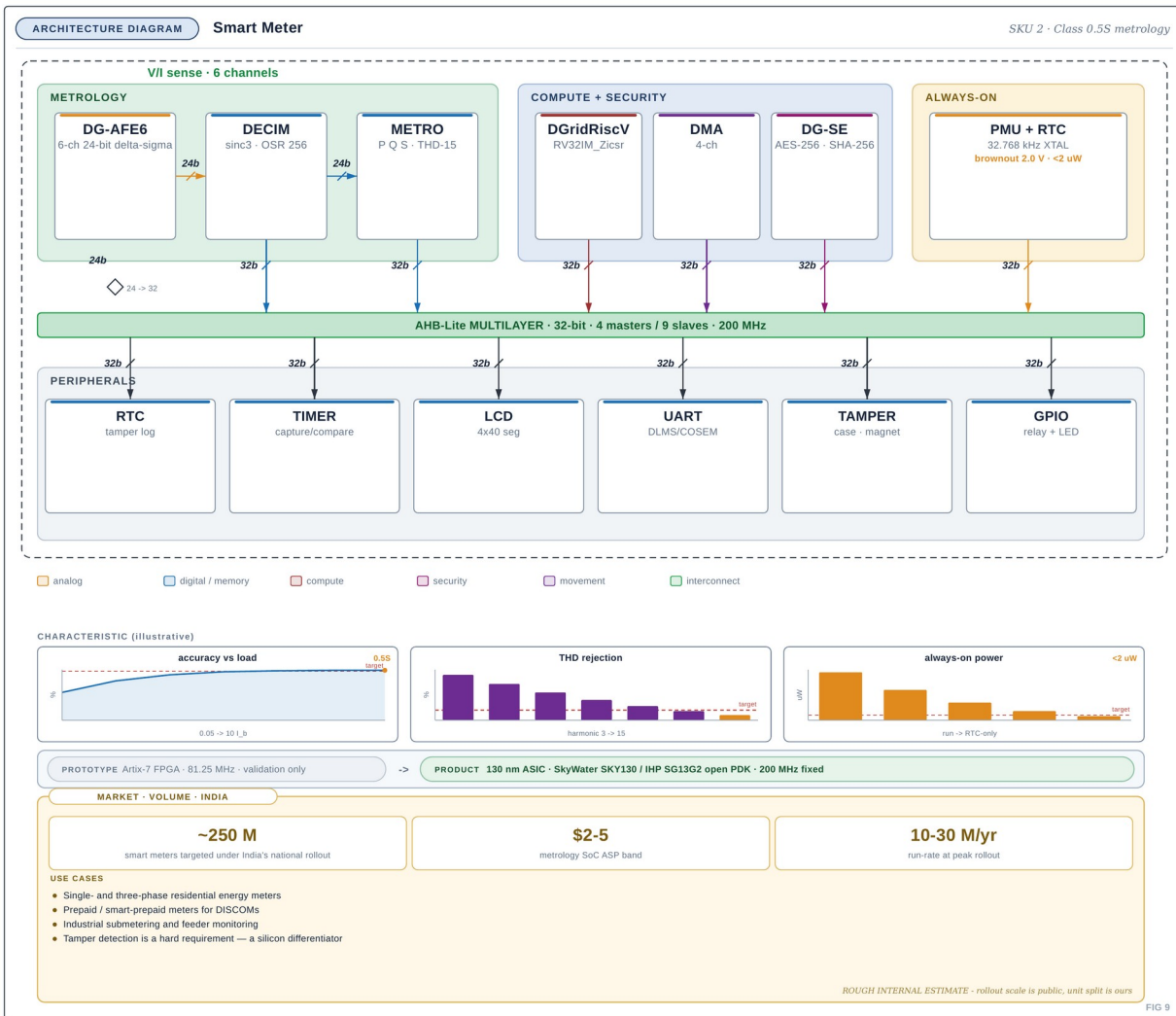
FPGA prototype (Artix-7, 81.25 MHz) = validation only; product is the 130 nm ASIC at 200 MHz. ReRAM via sky130 experimental module, SCL eNVM fallback. First MPW: Cycle 1.



Smart-Meter SoC — Class 0.5S Metrology

SKU - 2

DG-AFE6 six-channel 24-bit $\Sigma\Delta$ · sinc³ OSR-256 · P/Q/S + THD-15 engine · AES/SHA security · <2 μ W always-on RTC domain.



Replaces

ADE9153/V9203-class AFE + separate meter MCU — one die with multilayer AHB (4 masters / 9 slaves) and tamper-backed RTC.

Socket & policy hook

~250M-meter national rollout, 10-30M/yr at peak; \$2-5 ASP band. Tamper detection is a tender requirement — a silicon differentiator, not a feature.

Status & node path

FPGA-validated blocks; 130 nm ASIC @ 200 MHz. eFlash constraint: ROM + ECC SRAM + encrypted QSPI; SCL eNVM variant for sealed meters. Cycle-1 MPW alongside SKU-1.

Hi-Rel PMIC — 28 V Avionics Rail Tree

SKU - 3

DO-160 / MIL-STD-461 front end · peak-current-mode pre-buck · four sequenced rails with foldback · SEU-hardened FSM (DICE + TMR).

Replaces

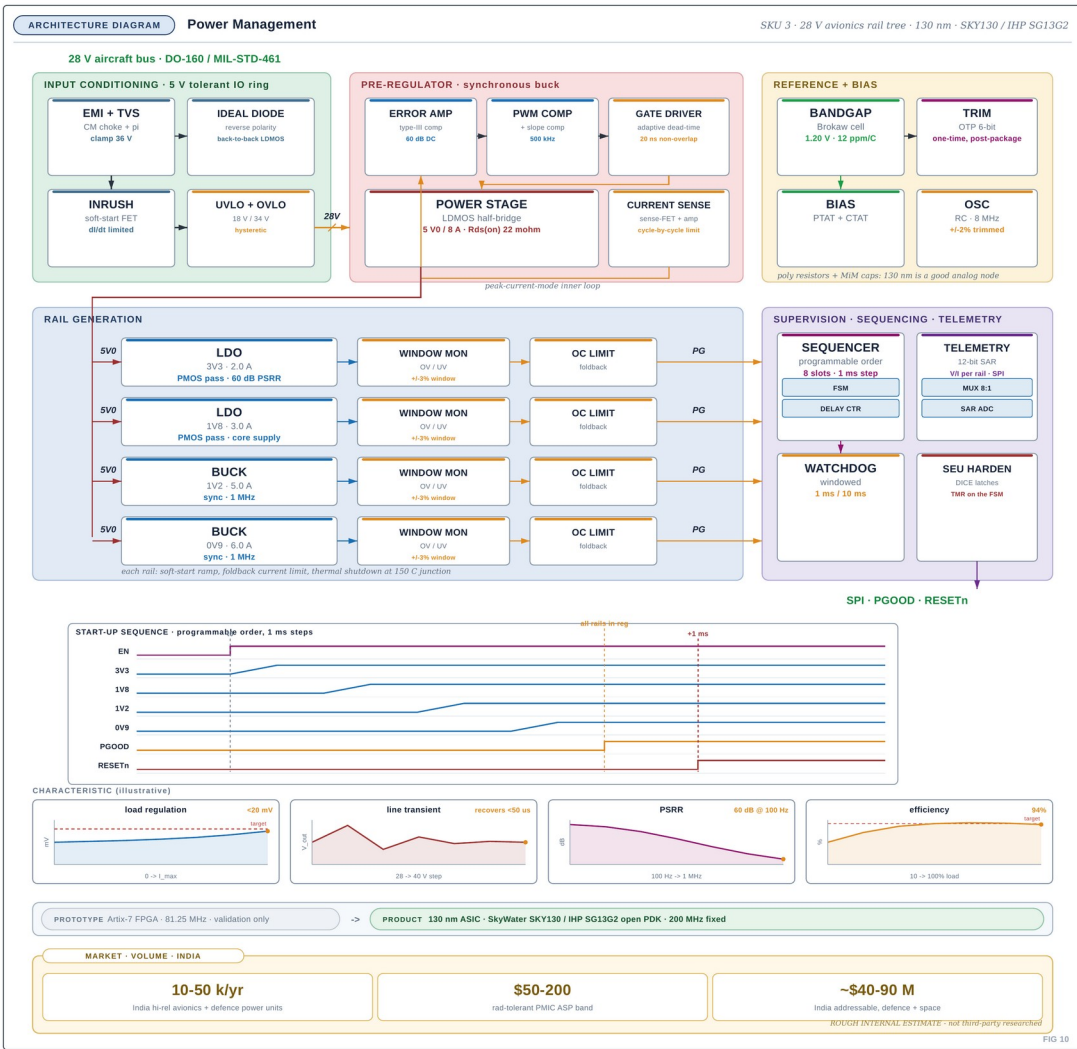
TI/ADI QML power sockets in every avionics and vetronics LRU — Brokaw bandgap 12 ppm/°C, type-III compensation, telemetry SAR — real analog design, not a digital wrapper.

Socket & policy hook

10–50K units/yr India hi-rel power at \$50–200 screened ASP — small volume, 60–75% GM, AVL-locked for decades once listed. SRIJAN NSG-5962 class.

Status & node path

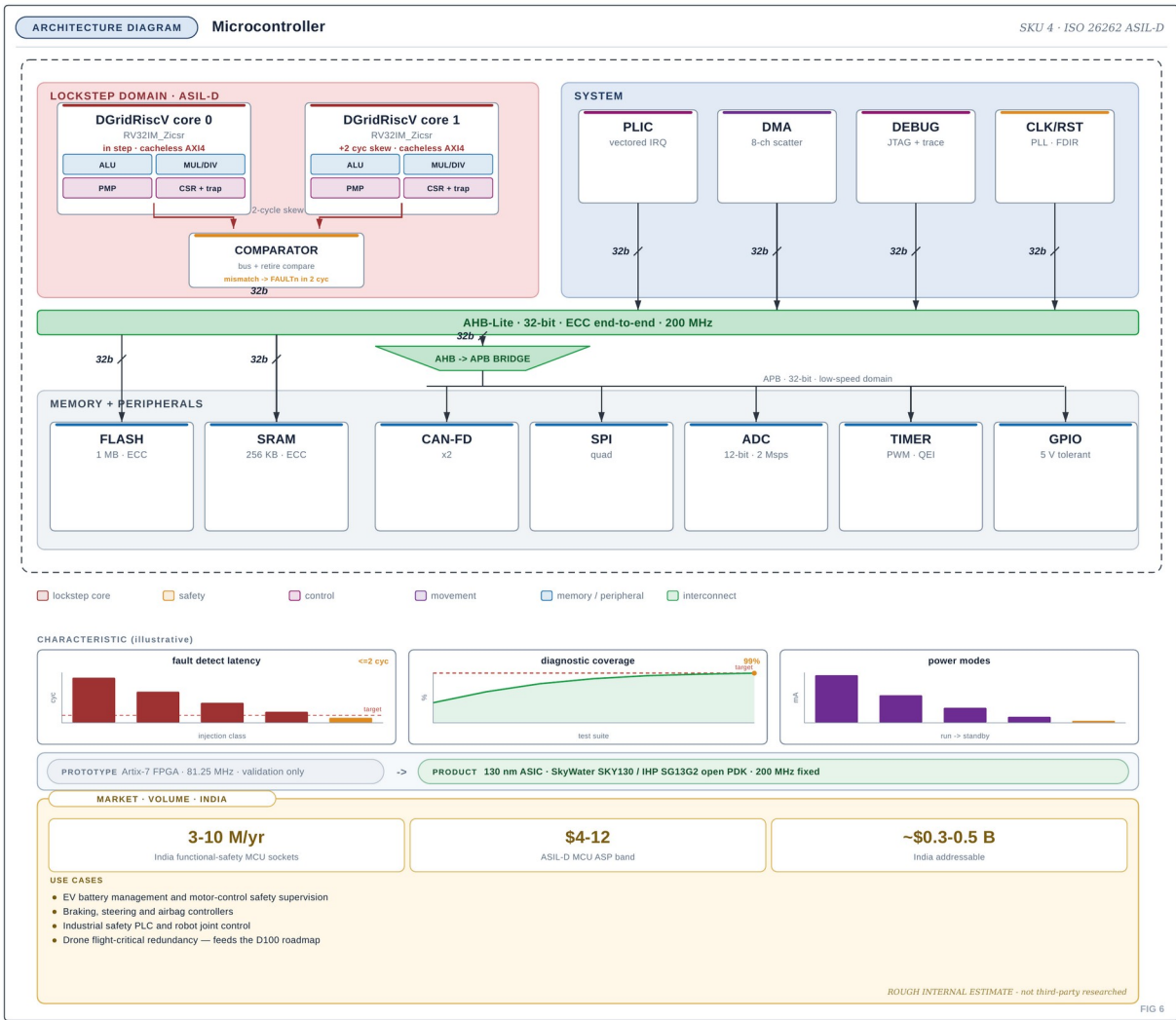
sky130 20V devices → SCL 180 nm production. 883 screening flow; SEU hardening by design (DICE latches, TMR on the sequencer FSM) — no rad-hard claim beyond that.



Lockstep MCU — ISO 26262 ASIL-D Path

SKU - 4

Dual DGridRiscV in 2-cycle skew · bus+retire comparator → FAULTn in 2 cycles · ECC end-to-end AHB · the FI-verified core productised.



Replaces

Microchip/Renesas functional-safety MCU sockets — 3–10M/yr India, \$4–12 ASP. The lockstep RTL already exists and is fault-injection tested from the D100 programme.

Socket & policy hook

EV BMS and motor-safety supervision, braking/steering controllers, robot joints — plus the FOC brain beside SKU-1 and flight-critical redundancy feeding the D100 roadmap.

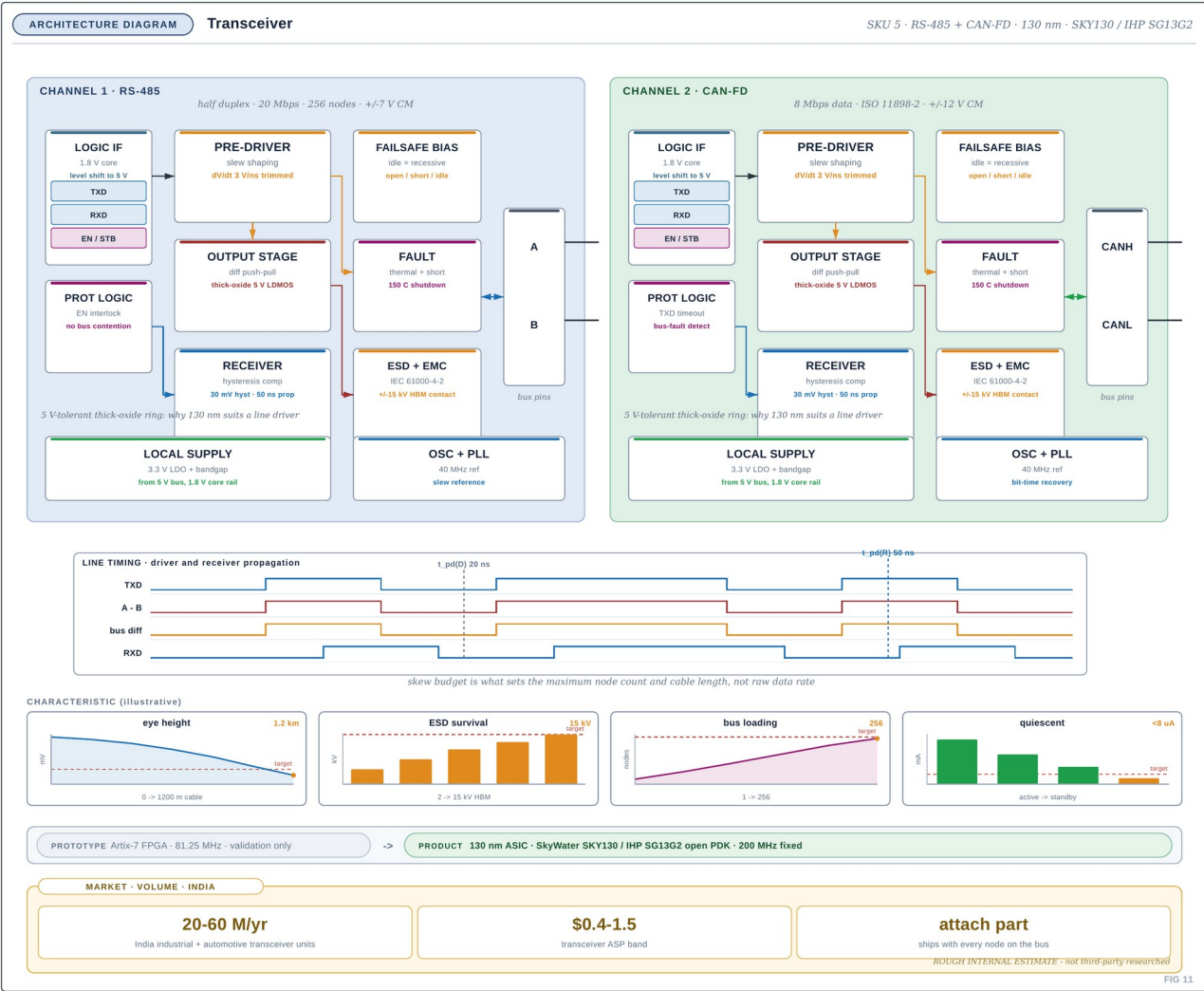
Status & node path

FPGA 81.25 MHz validation → 130 nm ASIC 200 MHz. Flash shown (1 MB ECC) maps to SCL eNVM variant; open-PDK v1 uses ROM+SRAM+XTS-QSPI — stated per canon.

Interface Transceiver — RS-485 + CAN-FD

SKU - 5

Thick-oxide 5 V LDMOS output stages • 30 mV hysteresis receivers • failsafe idle-recessive bias • ± 15 kV HBM • line-timing budget shown.



Replaces

TI/ADI/Renesas interface parts — the classic obsolescence + FSC-5962 sourcing-bar wedge. 20-60M/yr attach-part volume: ships with every node on every bus.

Why 130 nm wins here

The sheet says it: a 5V-tolerant thick-oxide ring is exactly what a line driver wants — advanced nodes are actively worse for this part class.

Status & node path

Skew budget — not raw bitrate — sets node count and cable length; designed accordingly. $\pm$ CM per ISO 11898-2; HV bus-fault variant on SCL/IHP. Cycle-2 MPW.

Voltage Supervisor — Quad-Rail, Latched Fault

SKU - 6

Chopper-stabilised comparators · 0.1%-matched poly ladders · 8 μ s deglitch · curvature-corrected bandgap 10 ppm/°C · windowed WDT.

Replaces

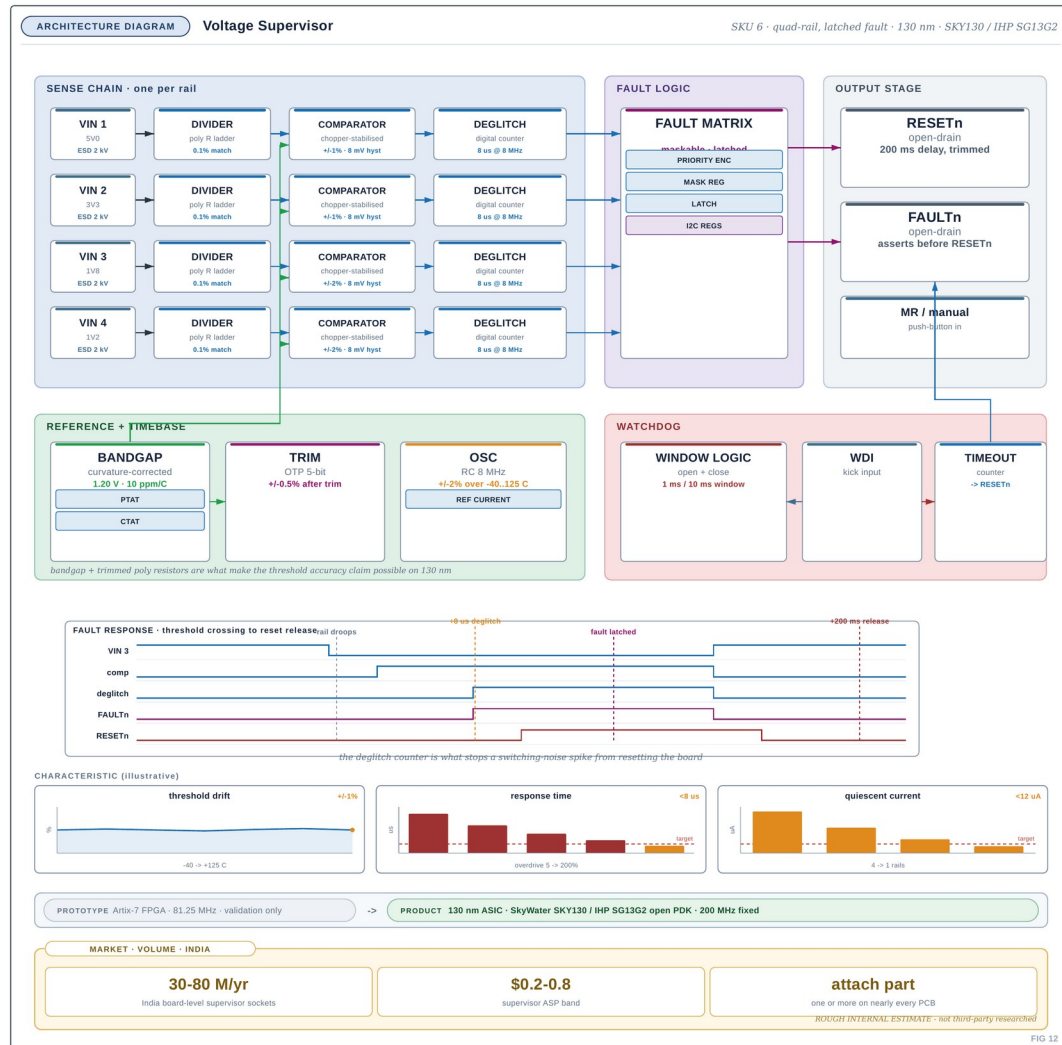
TI/Maxim supervisor sockets — 30–80M/yr attach volume, one or more on nearly every PCB; \$0.2–0.8 ASP commercial, screened grades 10–100 $\times$.

Qualification pathfinder

Simplest die in the catalogue → first through MIL-883 — carries the screening flow for everything behind it while filling the highest-socket-count position (every LRU, every D100 module).

Status & node path

The deglitch counter is the design story: switching-noise immunity without slowing real fault response (<8 μ s). sky130 → SCL. Cycle-1 or -2 MPW.



77 GHz MIMO Radar — SiGe Front End + CMOS Baseband

SKU-7

2TX/4RX · IHP SG13G2 HBTs (ft 350 GHz) · 4 GHz sweep → 3.75 cm resolution · range/Doppler FFT + CFAR + beamforming on the CMOS die.

The die boundary

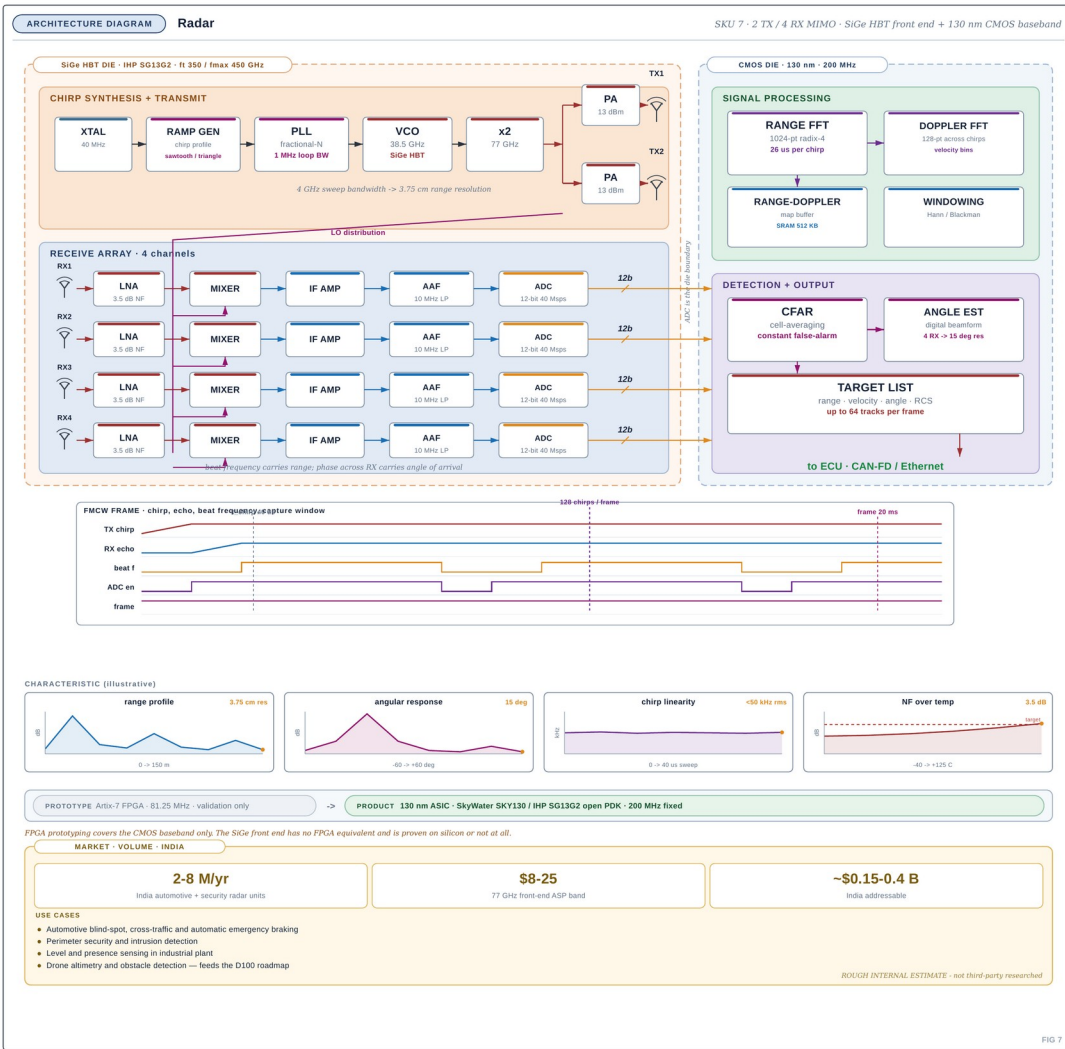
Stated on the sheet: ADC is the boundary — SiGe HBT die (chirp synth, PAs, 4 LNA/mixer/IF chains) + 130 nm CMOS die (12-bit 40 Mps ADCs, FFTs, CFAR, tracker). Gen-1 automotive partitioning.

Socket & policy hook

AD2 mirror-tower radar (captive) + defence perimeter/counter-UAS + automotive AEB — the only SKU with simultaneous captive automotive and defence sockets.

Honesty on the sheet

“The SiGe front end has no FPGA equivalent and is proven on silicon or not at all” — the risk statement is printed on the figure itself. IHP MPW FY28.



Rugged Display Driver — TFT Source Driver + TCON

SKU - 8

LVDS/DSI in · SXGA @ 60 Hz · 1280 × 10-bit column DACs · 0–12 V HV amps · temp-compensated gamma · frame-freeze BIST (ASIL-B capable).

The policy anchor

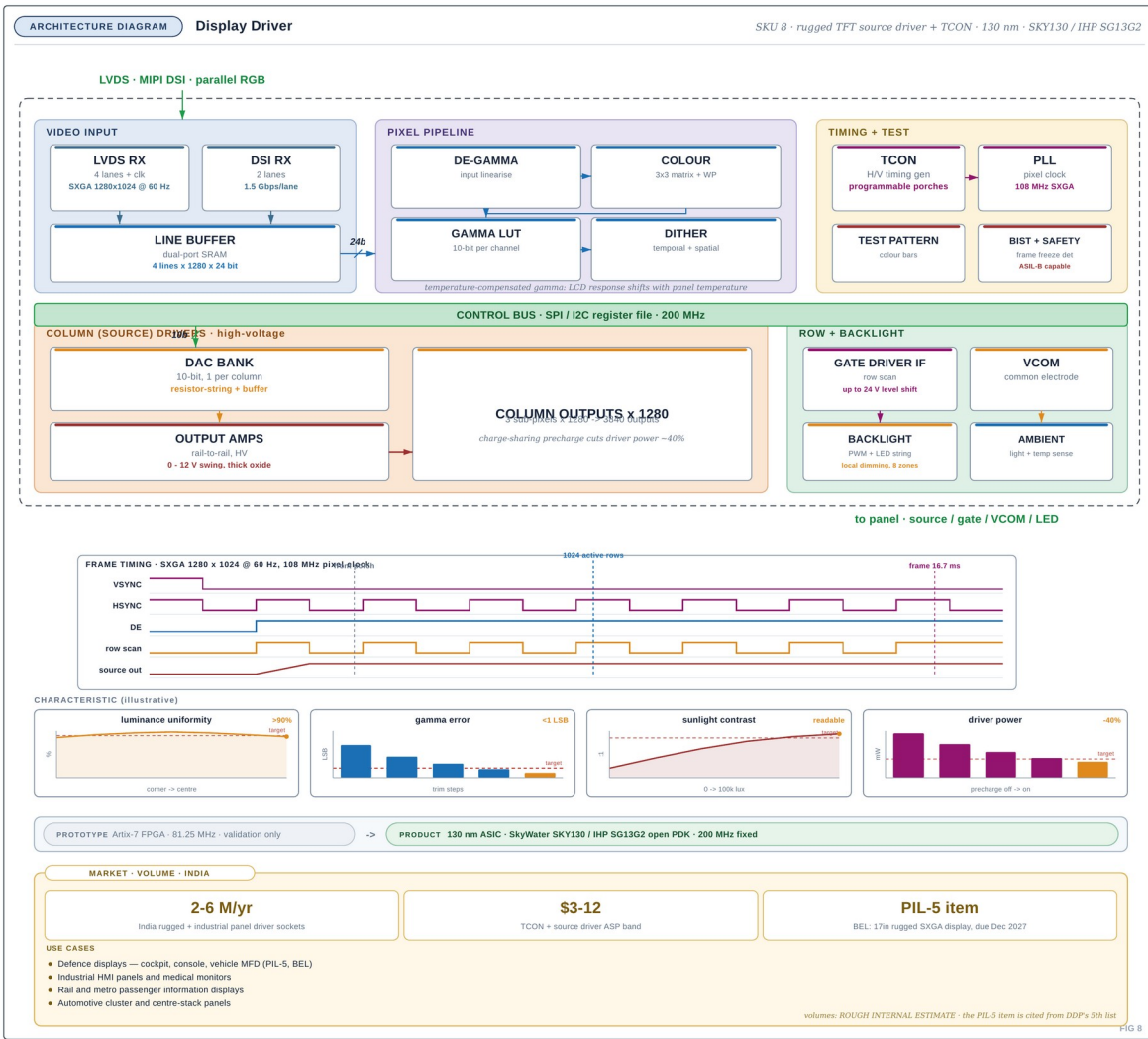
PIL-5 line item: BEL 17-inch rugged SXGA display, due Dec 2027 — a named DPSU socket with a date, cited from DDP's 5th list on the sheet itself.

Replaces

Novatek/Himax-class TCON + source-driver imports in defence MFDs, industrial HMI, rail PIS and automotive clusters — 2–6M/yr sockets at \$3–12.

Why 130 nm wins

HV output amps (0–12 V swing, thick oxide) and 24 V gate-driver level shift are mature-node territory; charge-sharing precharge cuts driver power ~40%. Cycle-3 MPW.



SDV Zonal Controller + Gateway

SKU - 9

ASIL-D lockstep island • EVITA-full HSM + secure boot/OTA • TSN switch (802.1Qbv/AS) • CAN-FD×8, CAN-XL, LIN×8, FlexRay • smart fuse ×16.

Scope honesty (on-sheet)

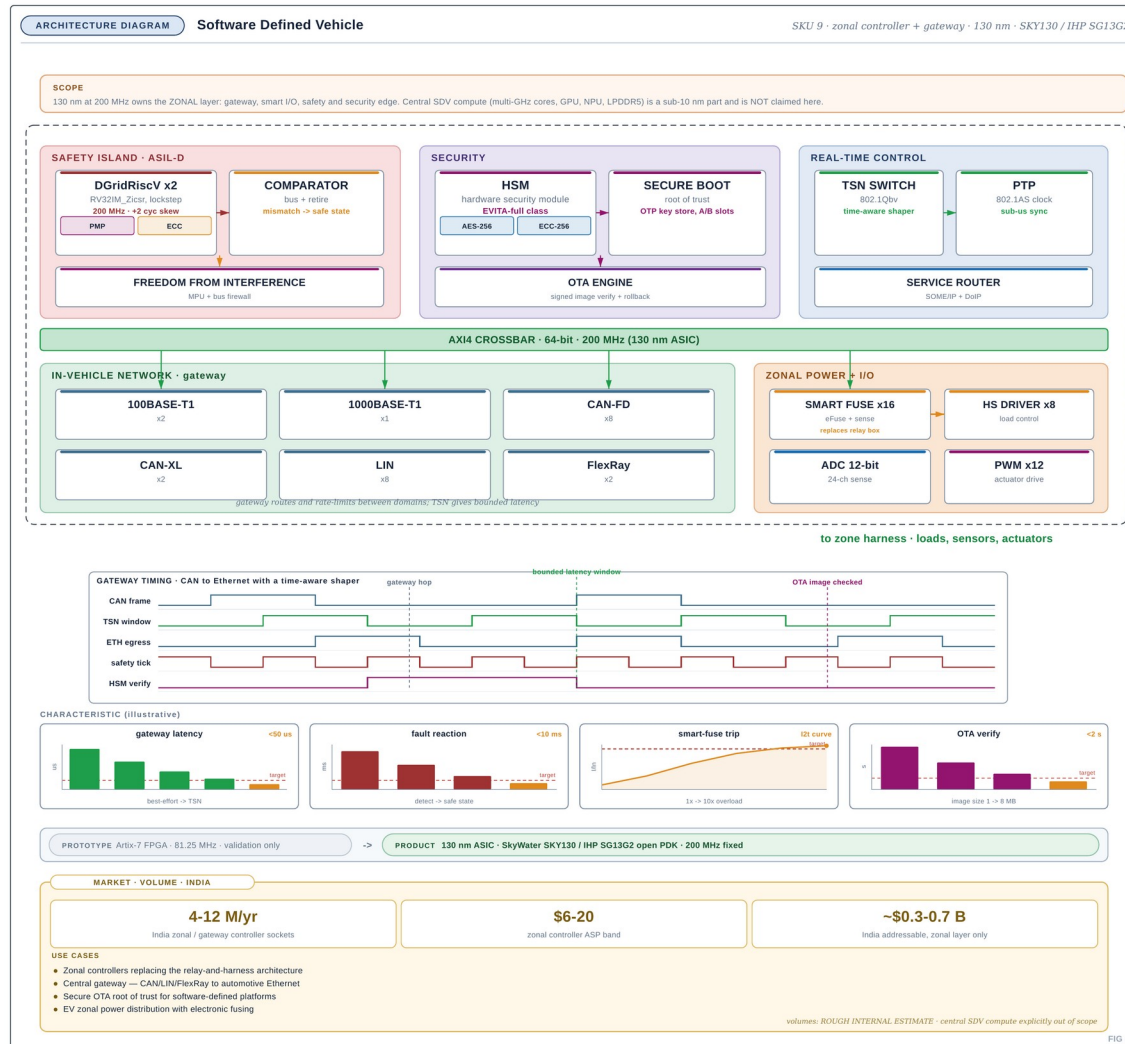
130 nm at 200 MHz owns the ZONAL layer — gateway, smart I/O, safety and security edge. Central SDV compute (multi-GHz, GPU/NPU, LPDDR5) is sub-10 nm and explicitly NOT claimed.

Socket & market

4–12M/yr India zonal sockets at \$6–20 as OEMs replace relay-and-harness architectures; ~\$0.3–0.7B addressable, zonal layer only — flagged internal estimate.

Where it fits

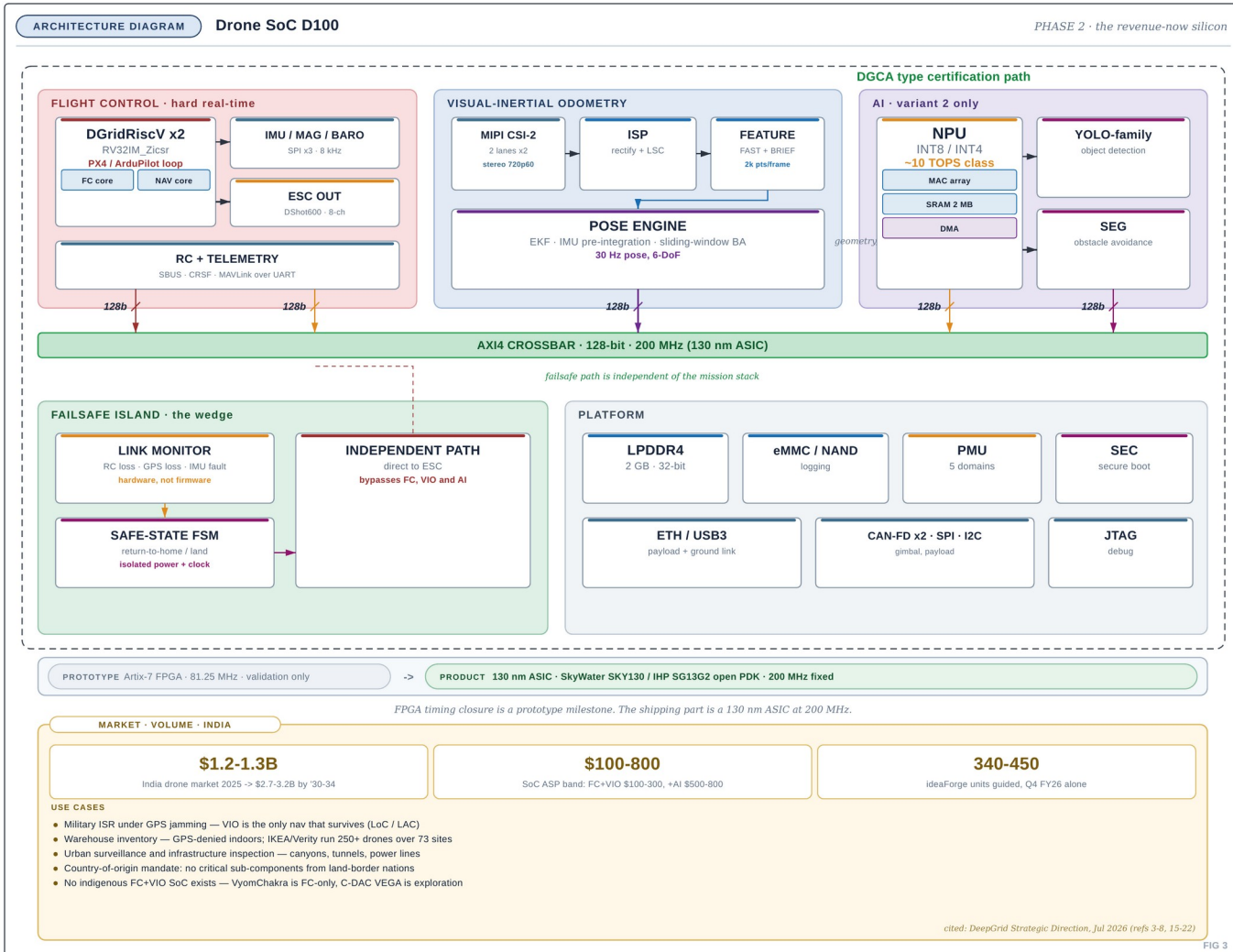
Bridges the CV ADAS heritage (AD2) and the DG SDV platform: this is the mature-node edge of the same architecture Design.png shows at the centre.



Drone SoC D100 — Phase 2, the Revenue-Now Silicon

D100

Flight control (PX4/ArduPilot loop) + visual-inertial odometry (EKF pose 30 Hz) + optional 10-TOPS NPU — with a hardware failsafe island independent of the mission stack.



The wedge (on-sheet)

Failsafe island in hardware, not firmware: link monitor → safe-state FSM → independent path direct to ESC, isolated power and clock — bypasses FC, VIO and AI. DGCA type-certification path.

Market (sheet, cited)

India drone market \$1.2–1.3B → \$2.7–3.2B by '30–34; SoC ASP \$100–800; ideaForge guided 340–450 units in Q4 FY26 alone. No indigenous FC+VIO SoC exists — gap stated with references.

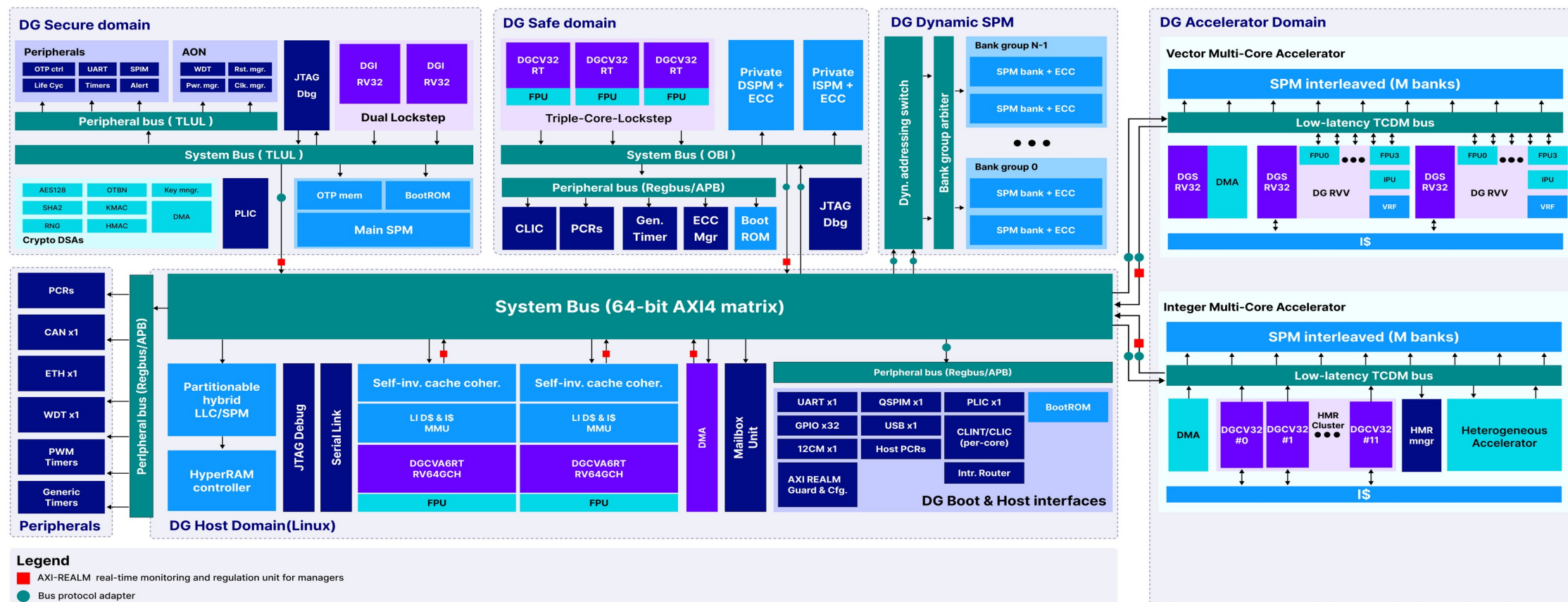
Status

FPGA prototype validates; product is 130 nm at 200 MHz fixed. VIO is geometry, not learned perception — survives GPS jamming (LoC/LAC use case).

DG SDV Platform — The Architecture Behind It All

TRACK B

Deepgrid's platform reference: secure + safe (triple-lockstep) + Linux host (RV64GCH) + vector & integer accelerator domains on a 64-bit AXI4 matrix with AXI-REALM QoS.



SKU-9 is the mature-node zonal edge of this platform; the D100 is its airborne sibling; SKU-3 powers, SKU-6 supervises, SKU-5 connects — one architecture, many dies.

SiP Integration — How the Dies Compose

SiP

Wire-bonded mature dies + one flip-chip 28 nm compute die on a 4-layer organic substrate — mature-node packaging at mature-node cost.

SiP · six mature dies + 28 nm compute on one organic substrate

= SDV domain controller / D100 UAV compute module

wire-bond ×6 + flip-chip ×1 — SiP-class multi-die, deliberately not UCle

ie: SPI / UART / GPIO over substrate · power tree from



Package summary

4-layer organic substrate · BGA 15×15 · wire-bond ×6 + flip-chip ×1 · die-to-die SPI/UART/GPIO over substrate · power tree from SKU-3 · SKU-6 supervises the package · deliberately not UCle.

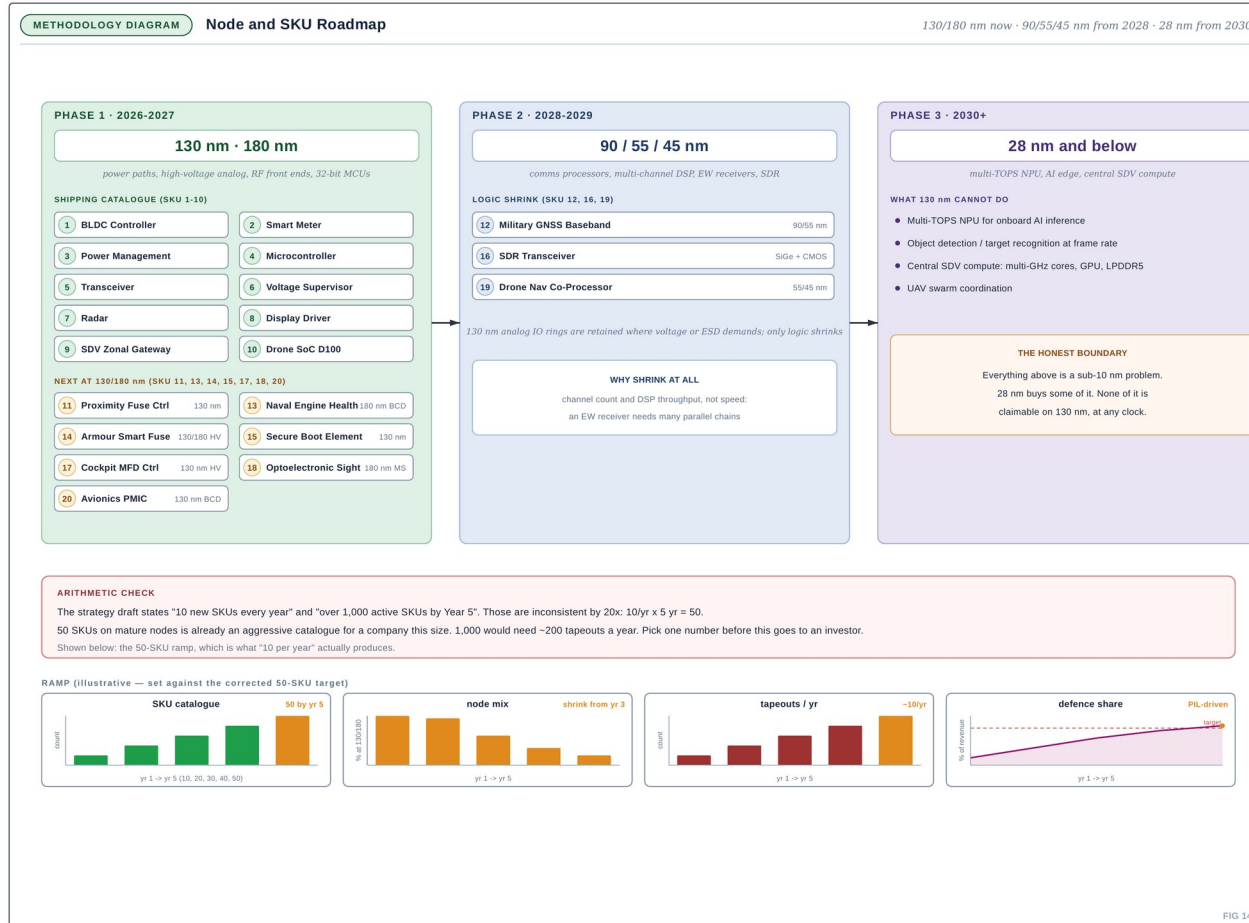
Wave 2

The 28 nm AI-compute die (D100 core) joins after the ₹50 Cr round — same substrate, same supervision, one SDV/UAV module.

Node & SKU Roadmap — Three Phases, Arithmetic-Checked

ROADMAP

130/180 nm ships the catalogue; 90/55/45 nm takes the logic-bound parts from 2028; 28 nm takes compute from 2030 — with the honest boundary and the SKU-ramp correction printed on the figure.



Phase 2 · why shrink at all

Channel count and DSP throughput — not speed vanity: GNSS baseband (90/55), SDR transceiver (SiGe+CMOS), drone nav co-processor (55/45). Analog IO rings stay 130 nm where voltage and ESD demand it.

Phase 3 · the honest boundary

"Everything above is a sub-10 nm problem. 28 nm buys some of it. None of it is claimable on 130 nm, at any clock." — printed on the roadmap itself.

The arithmetic check

The figure corrects its own draft: 10 SKUs/yr and 1,000 SKUs by Year 5 were inconsistent by 20x. Shown ramp: ~50-SKU catalogue by Year 5 — one number, before an investor finds two.